Number	Quantity	Board Designator	Description	Mouser Part Number	Other
1	2	C1, C2	0.01uF 50V 1206 SMD Capacitor	C1206C104K5RACTU	
2	2	D1, D2, D3, D4	1N4148W SMD SOD123	637-1N4148W	
3	4	R1, R2,R3,R4	2.2K Ohm 1206 SMD Resistor	652-CR1206FX-2201ELF	
4	2	R5, R6	300 Ohm 1206 SMD Resistor	652-CR1206JW-331ELF	
5	2	U1, U2	MCP23017SO SOIC-28	579-MCP23017-E/SO	
6	1	J1	IDC 1x20 Male Headder		TaydaElectronics.com
7	2	J2, J3	IDC 1x2 Male Headder		TaydaElectronics.com
8	6	J4, J5, J6, J7, J8, J9	Soldered Jumpers		Buss wire
9	3	Encoders_1/2, Encoders_3/4, Front_Panel	IDC 2x5 Male Box Headder		TaydaElectronics.com
10A	4	Encoders	PEC11R4220F-S0024	652-PEC11R-4220F-S24	
10B	2	Dual Decoders			

NOTE: The encoder type is a matter of personal preference. There are many possible choices here. Do your research.

NOTE: Can use four single encoders, two dual ecoders, or one dual encoder and two single encoders.